

ReClaimIt – The Lost and Found App
Add-On Intelligent Modules

Module 1: AI Enhancement to Existing Lost & Found System

This module manages the AI-based processing and matching of lost and found items after the form submission.

AI-Based Cloud Image Processing

When a user uploads an image, the process is handled through the AI layer:

- The image is uploaded to Cloudinary cloud storage.
- Cloudinary generates a secure image URL.
- The AI system retrieves and processes the image through this URL.
- Image features are extracted and analyzed by AI for matching and validation.

All image-related checking and processing are controlled by the AI system.

AI-Based Analysis and Matching

The AI system performs:

- Image feature extraction (color, object patterns, visual similarity)
- Cross-matching between Lost and Found database

A similarity score is calculated. If it exceeds a defined threshold, the system identifies a potential match.

Module 2: Bilingual Chat with Voice-to-Text Conversion

This module improves communication between users.

- Users can send voice messages.
- Voice messages are automatically converted into text.
- The system supports both Hindi and English.
- If needed, automatic translation is applied.
- Messages are displayed in chat inbox for smooth communication.

This removes language barriers and improves user experience.

Module 3: AI-Based Model for the Detection Of AI-Generated Images (Reserve)

This module ensures authenticity of Provided image.

Every image contains metadata, which includes:

- Source of the image
- Device or software used
- Date and time
- Editing details

The AI analysis :

- Image metadata
- Pixel structure and digital patterns

If an image is AI-generated or manipulated, its metadata or internal structure may show irregularities. The system detects these differences and classifies the image as real or fake.

Module 4: Future Scope – Realtime Institutional Integration (Research)

This module is currently under research.

In the future, the system can be integrated with:

- Universities
- Corporate offices
- Public institutions

This will allow real-time lost and found tracking within campuses and organizations. It will improve recovery speed and create a centralized lost-and-found ecosystem.